

MCG

Model Candidate Generator

EVA Machine Learning Reasoning Layer — Module Specification

Version 1.0 | References: EVA-ML-Master.docx

1. Purpose

The Model Candidate Generator determines which learning approaches EVA should consider for the defined prediction problem. Instead of blindly testing many algorithms, this module reasons about what types of models are appropriate given the problem type, dataset size, feature characteristics, user interpretability requirements, and stability needs.

Core Principle

EVA optimizes decisions, not leaderboard performance. The goal is a reliable, understandable model that supports the user's decision objective — not the most complex model available.

2. Inputs from Global Analysis Ledger

- `model_definition_record` — problem type, learning view schema, split strategy, class imbalance flag
- `user_intent_record` — stakeholder type, interpretability requirement, decision objective
- `hypotheses` — prior analytical findings about feature relationships

3. Candidate Selection Strategy

The module selects modeling families, not specific implementations. Selection depends on four factors: data size, feature structure, noise level, and explanation requirement.

The module creates a balanced portfolio including an interpretable baseline, a flexible learner, and a robustness-oriented method. This ensures meaningful comparison rather than random experimentation.

3.1 Candidate Count

The module outputs between 2 and 4 modeling approaches. If fewer than 2 approaches survive pre-selection filters, the module logs a warning in the `session_log` and proceeds with a single candidate only if it passes all downstream reliability checks. It must not silently reduce scope.

4. Interpretable Baseline Requirement

Every modeling session must include at least one interpretable candidate. This is mandatory and cannot be skipped.

- Provides a reference performance benchmark
- Provides understandable feature relationships for hypothesis checking
- Acts as a sanity check against complex models

If a complex model outperforms the baseline only marginally, the baseline is preferred per the Master PRD's reliability-over-complexity principle.

5. Complexity Control

5.1 Prefer Simpler Models When

- Dataset is small
- Features are few
- Noise level is high
- User requires explanation

5.2 Allow Higher Flexibility When

- Large dataset exists
- Complex nonlinear patterns have been detected
- Predictive accuracy is critical and stakeholder type permits reduced interpretability

6. Metric Selection

Evaluation metrics are chosen based on user intent and must be recorded with justification. Defaults by use case:

User Intent	Metric Guidance
Risk detection	Minimize missed risky cases — prioritize recall
Decision automation	Minimize false actions — prioritize precision
Estimation tasks	Minimize prediction deviation — prioritize RMSE or MAE
High interpretability requirement	Emphasize explainability; accuracy is secondary
Class imbalance flagged	Use F1, precision/recall, or AUC-ROC. Never use raw accuracy.

7. Hypothesis Alignment Check

The module compares candidate model assumptions with previously generated hypotheses from the GAL. If hypotheses suggest strong linear relationships, simpler models are prioritized. If relationships appear nonlinear or interaction-heavy, flexible candidates are included. Modeling assumptions that contradict observed data behavior are not permitted.

8. Output: candidate_model_record

Field	Contents
Candidate List	2–4 selected learning approaches
Reasoning	Why each candidate was chosen, referencing dataset characteristics
Expected Behavior	What patterns each candidate should capture
Expected Weaknesses	Known limitations of each candidate
Chosen Evaluation Metric	How success will be measured, with justification
Interpretability Expectation	Level of explanation possible for each candidate

9. Operational Rules

1. At least one interpretable candidate is mandatory in every session.
2. Candidate count must be between 2 and 4. Log a warning if fewer than 2 survive.
3. Candidate reasoning must reference dataset characteristics from the `model_definition_record`.
4. Metric choice must align with user intent and be justified.
5. Candidates cannot use modeling assumptions that contradict hypothesis findings.
6. Class imbalance flag must influence metric selection if present.

10. Relationship to Other Modules

Module	Relationship
PFTDC	Provides <code>model_definition_record</code> including problem type and class imbalance flag
TEM	Trains and evaluates the candidate set
RVEM	Verifies candidate behavior against hypotheses
Master PRD	Defines reliability-over-complexity preference and global rules